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GroLink: A general application programming interface for the plant-modeling platform GroIMP

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Summary

GroIMP (Kniemeyer, 2008), a graph-based plant modeling platform, is broadly used by students and researchers. Its integrated design, keeping modeling, simulation and analysis in the same graphical user interface, makes installation, learning and modeling easy. However, as data and computing power increased, so did the complexity of models, driving the need for integration into external tools and remote execution. The GroLink project (Oberländer, 2023) created a general HTTP API with additional client libraries to answer this need. Using this approach, it is possible to run and interact with GroIMP models from other software or devices using an extendable set of generalized commands. In addition, a number of examples have been created to demonstrate the potential of this approach.

Statement of need

GroIMP projects as well as the platform itself and the general usage of computers have evolved since GroIMP was first released 20 years ago. In some cases, the focus of projects shifted towards the integration of GroIMP into bigger pipelines or, due to the size and complexity of simulations, more performant executions were needed. From our experience with users and projects, we derived four main directions in the fields of automatization and integration to focus on.

GroIMP as a source for 3D models

The creation of 3D models, as (intermediate) results of simulations, is a key feature of GroIMP. While the export of the scene via the GUI works at any time, the direct export to another software is currently only possible after the end of the simulation (Lanwert, 2008) or via non-interactive file-based approaches. An addition would not only be interesting for live integration (as seen on other platforms (Fabrika, 2021)), but also for forwarding structural data (Zhu et al., 2021).
To improve this, it is necessary to enable other software to **execute simulation steps** of the model and to **export the scene at any time**.

Custom model interfaces

While GroIMP comes with a fully functional graphical user interface, there are cases where a simpler interface or additional features are needed. A well-explored way is to embed other software to create e.g. R-based charts (Gürtler, 2020) or videos of simulations (Leppich, 2020). Yet creating a new interactive interface for a GroIMP model is currently limited to either an implementation deep inside the platform (Smoleňová et al., 2012, 2013) or to an external

interface that restarts a non-interactive simulation in the background after each user input (Spehle, 2023). To enable realtime external model interfaces a **software independent interactive connection** would be required.

GroIMP as an external service

The idea of providing a file containing a model and set of instructions to GroIMP from another software was already introduced by the work of Qinqin Long (Long et al., 2018) on the co-simulation with the plant modeling platform OpenAlea (Pradal et al., 2008). Yet this work was focusing more on the theory of co-simulation and model transfer, therefore the integration follows a quite strict procedure. A more general integration as a service would need an approach that can work **independent from specific file formats or scenarios**.

Orchestration of multiple simulations

There are several scenarios where the automated execution of models or multiple versions of the same model is useful, for instance, the fitting of model parameters.

Currently, users are executing GroIMP in headless for each individual simulation using external scripts (Streit et al., 2016). This requires reading and writing data through external files since there is no interaction with the model and every model execution includes starting and stopping the platform. Moreover, the order of simulation steps has to be defined in the model, which requires additional implementation.

Most of these issues could be solved by using the already existing GroIMP HTTP server (Kniemeyer, 2008; Lanwert, 2008) (not the API server), yet this solution requires even more changes in the simulation code and seems to be too technically challenging. Therefore, an improved solution needs to allow users to easily execute several simulations on a running instance **in parallel, while being able to interact** with them **without additional implementation** on the simulation.

GroLink

The GroLink project created a GroIMP plugin that enables the software to run as an non-blocking HTTP server that provides a stateless API. The design uses the fact that in GroIMP, each project (workbench) runs independently on its own thread and that any interaction is done through defined commands.

For each request, the API server resolves the specified workbench and the referenced command and pushes this to the workbench for execution.

This design was chosen based on the needs discussed above:

- Software independence is given by the usage of HTTP, which is one of the most used communication protocols.
- Id-based addressing of workbenches allows for parallel interactive simulation.
- By using commands, a quite general access to GroIMP's functionalities is provided. These could even be extended in additional plugins, as explained in the wiki.
- Since this approach mimics the same kind of interaction the GUI uses, changes in the model are almost never required.

To simplify the usage of the GroLink API, we created additional libraries in Python(GroPy) and R(GroR) that can handle the connection to the server and the individual projects.

Installation and usage

The GroLink API-Server is available in the GroIMP plugin manager and can be installed directly from within GroIMP. To run the server, only the additional command-line argument "-a `API`" must be provided on startup. This is explained in more detail for the different platforms in the wiki.

The simplest way to send requests to the running API server is using a web browser, a tutorial for this can be found in the wiki and a list of all commands in the git repository of the plugin.

For the GroPy and GroR libraries, installation instructions and lists with the commands are provided in the repositories. Specific tutorials can be found in the wiki.

Examples

To explore the abilities of this new interface, we created three different client projects aiming in the directions we considered above. Additionally, two students' projects used the GroLink API to conduct parallelized sensitivity analysis, and first tests on a Kubernetes setup were done.

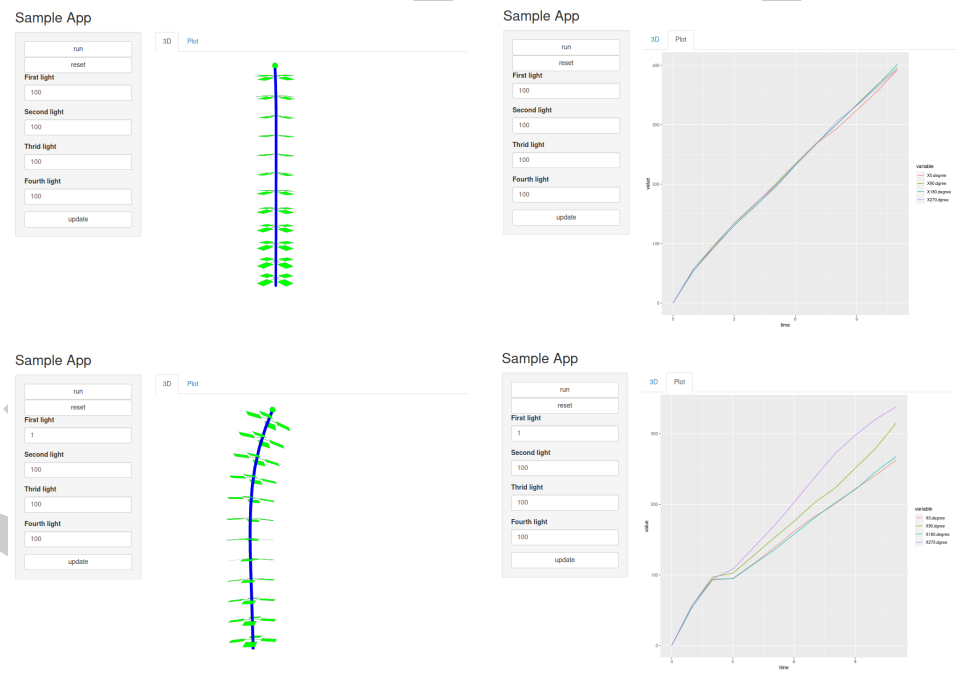

Figure 1: An interactive R-Shiny web interface to visualize a plant and manipulate its environment.

Figure 2: The interactive visualization of a GroIMP forestery scene in the Godot game engine.

1. To demonstrate the usage of import and export, the GroPy integration, and the possibilities for automation, we created an automated processing pipeline, that applies GroIMP's spectral ray-tracer on the imported 3d-model and returns a colored 3d-model with colors representing the radiation for each organ of the simulated plant.
2. For highlighting project based interaction, data transfer, the GroR integration and the ability to create custom UI's we implemented two interactive RShiny web applications: A generalized viewer for GroIMP models including interaction, and an interface to manipulate the environment of a specific plant model and observe the reactions (Figure 1).
3. Focused on the interaction with the 3D scene through live integration as well as the software independence of an HTTP API, different forest models were linked to a Godot game (Godot, n.d.) where a user can walk through the forest, cut or plant trees, and trigger growth steps which are then simulated in GroIMP (Figure 2).

Discussion

Considering the examples and students projects, GroLink seems to be suitable to answer the pointed out needs. Only regarding the orchestration we found that the old HTTP-Server still is a better fit, as long as no interaction with the orchestrated model is required during the simulation. This includes most cases of sensitivity analysis, but for more complex scenarios such as digital twinning, the GroLink approach would be the better option.

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